

Dieldrin-induced neurotoxicity: relevance to Parkinson's disease pathogenesis.



Parkinson's disease (PD) is increasingly recognized as a neurodegenerative disorder strongly associated with environmental chemical exposures. Recent epidemiological data demonstrate that environmental risk factors may play a dominant role as compared to genetic factors in the etiopathogenesis of idiopathic Parkinson's disease. Identification of key genetic defects such as alpha-synuclein and parkin mutations in PD also underscores the important role of genetic factors in the disease. Thus, understanding the interplay between genes and environment in PD may be critical to unlocking the mysteries of this 200-year-old neurodegenerative disease. Pesticides and metals are the most common classes of environmental chemicals that promote dopaminergic degeneration. The organochlorine pesticide dieldrin has been found in human PD postmortem brain tissues, suggesting that this pesticide has potential to promote nigral cell death. Though dieldrin has been banned, humans continue to be exposed to the pesticide through contaminated dairy products and meats due to the persistent accumulation of the pesticide in the environment. This review summarizes various neurotoxic studies conducted in both cell culture and animals models following dieldrin exposure and discusses their relevance to key pathological mechanisms associated with nigral dopaminergic degeneration including oxidative stress, mitochondrial dysfunction, protein aggregation, and apoptosis.